

Solid Oxide Electrolysis Oxygen Extraction Manual

1. Solid Oxide Electrolyzer Principles

Martian atmospheric carbon dioxide is compressed to 1.0 bar and heated to 800 degrees Celsius across yttria-stabilized zirconia (YSZ) catalytic membranes.

2. Catalytic Cleavage & Yield

Electrolysis splits CO₂ into oxygen ions and carbon monoxide byproduct: $2\text{CO}_2 \rightarrow 2\text{CO} + \text{O}_2$. Net breathable oxygen output reaches 22.0 grams per hour per stack module.

3. Gas Purity & Scrubbing

Produced oxygen passes through ceramic particulate filters and silver-zeolite scrubbers, certifying 99.8 percent medical-grade breathable purity prior to storage liquefaction.